

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
TSCS - Midterm Examination Spring 2026

Natural Language Processing (CT-485) – A

Max Marks: 20

Time: 90 minutes

Note: Attempt all questions.

Q.1) Explain Sentiment Analysis, Tokenization, and Lemmatization, discuss their importance in Natural Language Processing (NLP), and provide examples for each technique.

[CLO-1: 5 Marks]

Q.2) Classify the sentence “I enjoy NLP” as Positive or Negative using the Naive Bayes classifier, given the following training data. Show necessary steps.

Sentence	Sentiment
NLP is very interesting	Positive
I enjoy learning about machine learning	Positive
NLP is too difficult	Negative
I hate debugging NLP models	Negative

[CLO-2: 5 Marks]

Q.3) Analyze below the raw corpus sample:

"um I think NLP is... like very cool!!! but... u know it needs CLEANING??"

1. Identify at least four preprocessing challenges in this text.
2. Explain why addressing these challenges is important for building reliable NLP models.

[CLO-2: 5 Marks]

Q.4) Apply appropriate NER technique on the given sentences with BIO Tagging.

1. “Elon Musk announced a new project in San Francisco on Friday”.
2. “Professor Alan Turing presented a lecture on Artificial Intelligence at Cambridge University in 1950”.

[CLO-3: 5 Marks]